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Seeing direct and averted gaze activates the approach–avoidance motivational brain systems

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ABSTRACT

Gaze direction is known to be an important factor in regulating social interaction. Recent evidence suggests that direct and averted gaze can signal the sender's motivational tendencies of approach and avoidance, respectively. We aimed at determining whether seeing another person's direct vs. averted gaze has an influence on the observer's neural approach–avoidance responses. We also examined whether it would make a difference if the participants were looking at the face of a real person or a picture. Measurements of hemispheric asymmetry in the frontal electroencephalographic activity indicated that another person's direct gaze elicited a relative left-sided frontal EEG activation (indicative of a tendency to approach), whereas averted gaze activated right-sided asymmetry (indicative of avoidance). Skin conductance responses were larger to faces than to control objects and to direct relative to averted gaze, indicating that faces, in general, and faces with direct gaze, in particular, elicited more intense autonomic activation and strength of the motivational tendencies than did control stimuli. Gaze direction also influenced subjective ratings of emotional arousal and valence. However, all these effects were observed only when participants were facing a real person, not when looking at a picture of a face. This finding was suggested to be due to the motivational responses to gaze direction being activated in the context of enhanced self-awareness by the presence of another person. The present results, thus, provide direct evidence that eye contact and gaze aversion between two persons influence the neural mechanisms regulating basic motivational–emotional responses and differentially activate the motivational approach–avoidance brain systems.

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Eye gaze is a powerful stimulus in social interaction. Seeing another person looking at you is likely to indicate that he or she is attending to you, whereas another person's gaze averted away from you indicates that his or her attention is directed to somewhere else. Not surprisingly, people are rather accurate at discriminating whether another person is looking straight at them or whether the gaze is averted, especially when the other person's face is seen from straight ahead (Gamer & Hecht, 2007). Even new-born infants (2 days old) can discriminate between straight and averted gaze suggesting that the ability to detect gaze direction is innate (Farroni, Csibra, Simion, & Johnson, 2002). In all, gaze direction serves several important functions in complex social interaction processes such as regulation of interaction, facilitation of communicational goals, and expression of intimacy and social control (Kleinke, 1986).

In the course of evolution, the ability to use information conveyed by the gaze direction of conspecifics and predators may have offered biological advantage for the survival of the species, especially for the sub-human and human primates (Emery, 2000). Considering the social and biological significance of gaze direction it is not, perhaps, surprising that the visual system of the primate brains has been shown to be equipped with specialized mechanisms for detecting eyes and gaze direction (e.g., Allison, Puce, & McCarthy, 2000; Calder et al., 2007; Perrett, Hietanen, Oram, & Benson, 1992). Recently, the effects of another person's direct and averted gaze on face perception and visual cognition have received much attention. It has been demonstrated, for example, that direct gaze is detected more efficiently relative to averted gaze (Conty, Tijus, Hugueville, Coelho, & George, 2006; Senju, Hasegawa, & Tojo, 2005; Von Grünau & Anston, 1995), that direct gaze enhances facial gender-categorization and access to gender-related semantic information (Macrae, Hood, Milne, Rowe, & Mason, 2002), and that direct gaze improves implicit memory for faces (Mason, Hood, & Macrae, 2004; Vuilleumier, George, Lister, Armony, & Driver, 2005). Compatible with behavioral findings, neurophysiological and imaging studies have suggested that direct gaze induces enhanced visual processing of facial information (Conty, NíDiaye, Tijus, & George, 2007; George, Driver, & Dolan, 2001; Pelphrey, Viola, & McCarthy, 2004). Seeing another's direct gaze has also been shown

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to automatically activate brain regions implied in "mentalizing", attribution of mental states to self and others (Kampe, Frith, & Frith, 2003; Wicker, Perrett, Baron-Cohen, & Decety, 2003), and direct gaze by attractive faces increases brain activity in the striatal reward systems (Kampe, Frith, Dolan, & Frith, 2001). Averted gaze, in turn, has been shown to trigger automatic shifts of visual attention in the gazed-at direction (e.g., Driver et al., 1999; Friesen & Kingstone, 1998; Hietanen, 1999) and this type of attention orienting seems to be subserved by specialized neural mechanisms (Hietanen, Nummenmaa, Nyman, Parkkola, & Hämäläinen, 2006). Averted gaze has also been shown to activate medial prefrontal cortex, an area associated with "mentalizing" (Calder et al., 2002). Interestingly, averted gaze by attractive faces has been shown to decrease activity in the brain reward systems (Kampe et al., 2001).

Recent empirical evidence suggests that direct and averted gaze can signal the sender's motivational tendencies of approach and avoidance, respectively. Adams and Kleck (2003, 2005) investigated the effect of gaze direction on recognition of facial expressions and showed that when the behavioral intent of these two signals matched the recognition was enhanced. For example, the recognition of facial happiness and anger (both emotions associated with approach motivation) was faster when combined with direct rather than averted gaze, whereas the recognition of fearful and sad faces (both emotions associated with avoidance motivation) was faster when combined with averted than straight gaze. In the present study, we aimed at determining whether seeing another person's direct vs. averted gaze could have an influence on the observers at the level of their basic motivational–emotional responses. Because of a strong inclination to reciprocity and synchronization in social behavior as evidenced, for example, in the attunement of the facial expressions, the body posture, and the gestures between social interactants (Oberman & Ramachandran, 2007), we hypothesized that another person's direct and averted gaze would initiate in the observer respectively, motivational tendencies of approach and avoidance.

The systems regulating basic motivational–emotional responses comprise of sets of neural networks regulating (i) the *direction* of the responses, i.e., the motivational tendency to approach or avoid the source of stimulation and (ii) the energy used for these responses, i.e., the *intensity* of the motivational tendency (Lang, Bradley, & Cuthbert, 1990). To test the hypothesis that another person's direct and averted gaze would initiate respectively, the motivational tendencies of approach and avoidance, we measured hemispheric asymmetry in the frontal electroencephalogram (EEG) and skin conductance responses (SCR) to another person's direct and averted gaze. The relatively stronger activation of the left than the right frontal cortex has been associated with the activation of the approach-related motivational system, whereas the relatively greater activation of the right frontal cortex has been associated with the activation of the avoidance system (Davidson, 1984, 2004; Harmon-Jones, 2003, 2004; Harmon-Jones, Lueck, Fearn, & Harmon-Jones, 2006; Van Honk & Schutter, 2006). Skin conductance, in turn, reflecting the functioning of the sympathetic autonomic nervous system, has been considered a good index of the general energetic level (arousal) of behavior (Andreassi, 2000). Thus, by combining the measures of the frontal EEG asymmetry and the SCR, we aimed to demonstrate that another person's gaze direction has an effect on the functioning of the two neural systems postulated to be involved in the regulation of motivational tendencies. To complement the physiological measures, the participants also rated their subjective experiences of arousal and emotional valence (pleasantness–unpleasantness) to different stimuli after the physiological measurements.

Another question we examined was whether it would make a difference if the participants were looking at the face of a real person or a picture. Virtually all research on face perception is currently conducted by presenting pictures of faces on a computer screen. This allows a stringent control over the stimuli and accurate time-locking of behavioral and physiological responses to the stimuli. However, it is possible that not all the responses are identical when collected to pictures of faces vs. to a face of another person sitting live opposite to the observer. Intuitively and based on everyday experiences, this difference may become particularly evident when another person's gaze direction is concerned. When sitting in a café and reading a magazine, seeing a face on a page looking straight to you (to the camera) is likely to elicit a different reaction than when lowering your hands and seeing behind the magazine somebody in the next table looking directly at you. As Argyle (1981, p. 238) pointed out "being looked at by another can be interpreted as 'being observed' by that person" and "this leads to self-consciousness and concern with self-presentation." It is hard to imagine that viewing a picture of a face with a direct gaze will result in such processes, at least, not to the same extent as when facing a real person. Thus, by comparing responses to looking at a live person vs. a picture with each other, we aimed at determining whether the knowledge of being an object of another person's scrutiny has an effect on the responses in addition to that evoked by the visual information derived from the face and gaze direction.

In the present study, the participants viewed a face/gaze stimulus and a control object (a radio) in four different conditions factorially manipulating the gaze/object direction (direct and averted) and the stimulus–presentation mode (picture and live). The pictures were presented on a computer monitor, whereas the live stimuli were presented through a liquid crystal shutter. The timing of the stimulus presentation was the same in the two modes of stimulus presentation. Power in the alpha band EEG activity (8–13 Hz) recorded from the left and right frontal channels and SCR were analyzed during the stimulus presentation. In the context of EEG frontal asymmetries, power in the alpha band EEG activity has been shown to be inversely related to cortical activity (Davidson, Jackson, & Larson, 2000). Three main hypotheses were tested: (i) perceiving a direct gaze elicits a relative left-sided frontal EEG activation (i.e., less alpha band power on the left than right frontal recording site) indicative of a motivational tendency to approach, and perceiving an averted gaze elicits smaller left-sided asymmetry or even right-sided asymmetry indicative of avoidance, (ii) the SCR is larger to faces than to the control object, indicating, in general, a greater intensity of the autonomic activation and strength of the motivational tendencies elicited by the facial than control stimuli, and (iii) the hypothesized pattern of results was expected to be more prominent for the live faces than for the pictures of faces.

1. Materials and methods

1.1. Participants

The participants were 20 adults (12 females, mean age = 24.8 years, range 20–40 years) with normal or corrected-to-normal vision. An additional five participants were tested but excluded due to excessive artifacts ($n = 2$) or technical error ($n = 3$) in the electroencephalography. In one of the remaining 20 participants, the skin conductance recording was not successful due to technical problems. Thus, the skin conductance responses were analyzed from 19 participants. Informed, written consent was obtained from each participant.

1.2. Stimuli and experimental procedure

The facial stimuli were the faces of the two female experimenters collecting the data. The gaze was either direct or averted (left or right) and the faces bore a neutral expression. Two small portable radios were used as control stimuli and were positioned either direct to the camera/participant or turned to the left or right. Radios were selected as control objects because they are usual everyday objects and, like faces, relatively symmetrical. Both classes of stimuli were presented in two conditions of presentation mode: picture and live. For the picture condition, the models and the radios were pictured with a digital camera, and the pictures were

Fig. 1. Top: gaze and control object in the direct and averted conditions. Bottom: a view of a face stimulus as seen through the liquid crystal shutter (in a transparent state) from the participant's perspective.

presented on a computer screen. In the live condition, the stimuli were presented through a voltage sensitive liquid crystal shutter (LC-TEC Displays AB) fixed to a large white panel, and positioned on the table between the model and the participant (Fig. 1). In both the picture and live condition, the participants were seated at a distance of 70 cm from the computer screen/panel. In the live condition, the model was sitting on the other side of the table at a distance of 30 cm from the panel. The retinal size of the live stimuli matched the size of the stimuli on the computer screen. The faces measured approximately 8.0° and 11° and the radios measured 14° and 10° horizontally and vertically, respectively. Great care was taken in order to ensure that the stimuli were visually comparable in the live and picture conditions.

The stimuli were presented in four blocks: live face, live radio, picture of a face, and picture of a radio. Within a block, the participant saw a total of 12 trials. The presentation order of the direct (six trials) and averted trials (six trials) was pseudo-random (no more than two consecutive trials of the same type). Each block included a face of one model or one radio only. However, each participant saw the faces of both models and both radios, one in the live block and the other in the picture block. Each trial lasted for 5 s. The ISI varied randomly from 30 to 45 s. During the ISI, the shutter remained opaque and the computer screen was black. A short audio signal was presented through the speakers 5 s before the start of the next trial to direct the participant's attention to the shutter/computer screen and, in the live conditions, to prepare the model to the opening of the shutter. Both live and picture blocks were always presented adjacent to each other, but the order of the stimulus (face/control) as well as the order of presentation mode (live/picture) was counterbalanced across the participants. Stimulus presentation was controlled in both stimulus presentation mode conditions by Neuroscan Stim software running on a desktop computer.

On arrival to the laboratory, the participant was introduced to the laboratory and an informed consent was obtained. The two experimenters described the general procedure and mentioned that, in the experiment, physiological reactions would be measured while the participant saw, among other things, their faces: one on a computer screen and another so that the experimenter would be sitting on the other side of panel and shutter. After this, both experimenters prepared the participant for the physiological measurements. The participants were instructed to look at the stimuli while remaining relatively still during the trials. No task was required, except to watch the stimuli as naturally as possible. The four experimental blocks were presented with short pauses between them. Immediately after the experiment, the participants assessed their feelings towards each of the eight different stimuli (i.e., the direct and averted face/radio in the live and picture condition) on the dimensions of affective valence and arousal. The participants were given four sheets of paper (one for each stimulus × presentation mode condition) on which two sets of 9-point Self-Assessment Manikin (SAM) scales (Bradley & Lang, 1994) were drawn, one for "direct" stimulus and another for "averted" stimulus. Ratings regarding the different stimulus × presentation mode condition were performed in the same order as the stimulus blocks were presented. The participants were asked to recall how they felt during the presentation of different stimulus conditions, and accordingly complete the valence and arousal ratings (in this order) stimulus by stimulus.

During the experiment, one experimenter sat behind a screen in such a way that she was able to observe and make a record of the participant's possible body

movements and also, in the live face condition, the model's possible movements, facial actions, and eye-blinks. These records verified that the models remained completely motionless and expressionless, and there were only a few blinks (eight blinks, corresponding to 3.7% of the trials included in the final data analysis).

1.3. Acquisition of the physiological data

Continuous EEG was recorded from F3, F4, F7, F8, C3, C4, P3, P4, and Cz. The signal was referenced to linked ears. Horizontal and vertical eye movements were recorded. Gentle skin abrasion and electrode paste were used to reduce the electrode impedances below 5 kΩ. The EEG signal was amplified with SynAmps amplifiers with a gain of 5000 and a 1–200-Hz band-pass filter (50-Hz notch filter enabled). The continuous signal was digitized at 1000 Hz and stored on a computer disk for off-line analyses.

For the skin conductance measurements, the electrodes (Ag/AgCl) were coated with electrode gel and attached to the palmar surface of the medial phalanxes of the index and middle fingers on the participant's left hand, which had been cleaned with an antiseptic liquid and gently abraded with fine sandpaper. Power Lab 400 equipment was used to measure the skin conductance. Data collection was controlled by Power Lab Chart v3.6 computer program running on a Power Macintosh 7100/80 computer. The sampling rate was 100 Hz.

1.4. Data analysis

The continuous EEG signal was corrected for blink artifact using a regression-based blink reduction algorithm (Semlitsch, Anderer, Schuster, & Presslisch, 1986). Eye movements other than blinks and other visible artifacts were eliminated by visual inspection. Artifact-free EEG during the 5-s stimulus period was segmented to eight 1.024-ms epochs with 50% overlap between adjacent epochs. Spectral power was calculated for each epoch using Fast Fourier Transform (FFT) with a 10% Hanning taper. The power spectra obtained were averaged over all artifact-free epochs within each trial and over separate trials within each experimental condition. Trials with less than 50% artifact-free epochs were excluded from averaging. For average power spectra within each condition, power density values (μV^2) within the alpha band (8–13 Hz) were calculated and natural log-transformed to normalize the distributions. Asymmetry scores were calculated for electrode pairs at frontal (F8/F7, F4/F3), central (C4/C3), and parietal (P4/P3) scalp regions by subtracting the ln-transformed power density values for the left site from that for the right site (Allen, Coan, & Nazarian, 2004). The main data analysis was confined to the data measured from the electrode pair F4/F3. The motivational-affective effects on the frontal EEG asymmetry are most typically detected from these recording sites (Davidson, 1995). Analysis of the data from the other recording sites revealed that, in the present study, too, the effects investigated were most clearly observed in the measurements from the mid-frontal electrode pair F4/F3.

The skin conductance response was defined as a maximum change from the baseline level (at the stimulus onset) during a 4-s time period starting after 1 s from the stimulus onset till the end of the stimulus presentation. The data were averaged over the six trials in each condition for each participant. For the statistical analyses, a log transformation [log(SCR + 1)] was performed to normalize the data. Responses contaminated by participant's body movements or technical problems with the measurement were eliminated from subsequent analysis. On the basis of these criteria, 5.3% of the trials were eliminated.

In the results of the analyses of variance, we also report the effect size, η_p^2. When post hoc testing of a significant main effect included multiple comparisons, the significance levels were Bonferroni-corrected.

2. Results

The mean EEG asymmetry scores based on the measurements from a frontally located pair of electrodes are presented in Fig. 2. These data were first analyzed with a three-way analysis of variance (ANOVA, within-subjects) having stimulus (face vs. control object), direction (direct vs. averted), and presentation mode (live vs. picture) as factors. The ANOVA indicated the main effect of stimulus, $F(1, 19) = 5.2$, $p < .05$, $\eta_p^2 = .21$, as well as stimulus × direction, $F(1, 19) = 5.6$, $p < .05$, $\eta_p^2 = .23$, and stimulus × direction × presentation mode, $F(1, 19) = 8.3$, $p < .01$, $\eta_p^2 = .30$, interactions. When the effect of direction was separately analyzed for each stimulus × presentation mode condition, the results showed that, in the live face-condition, the EEG-asymmetry scores for the direct gaze and averted gaze-conditions differed significantly from each other, $t(19) = 2.7$, $p < .02$. Notably, the results indicated left-sided asymmetry for the direct gaze and right-sided asymmetry for the averted gaze. In the other conditions, the direction of the gaze or the direction of the control object had no effect on the EEG asymmetry scores.

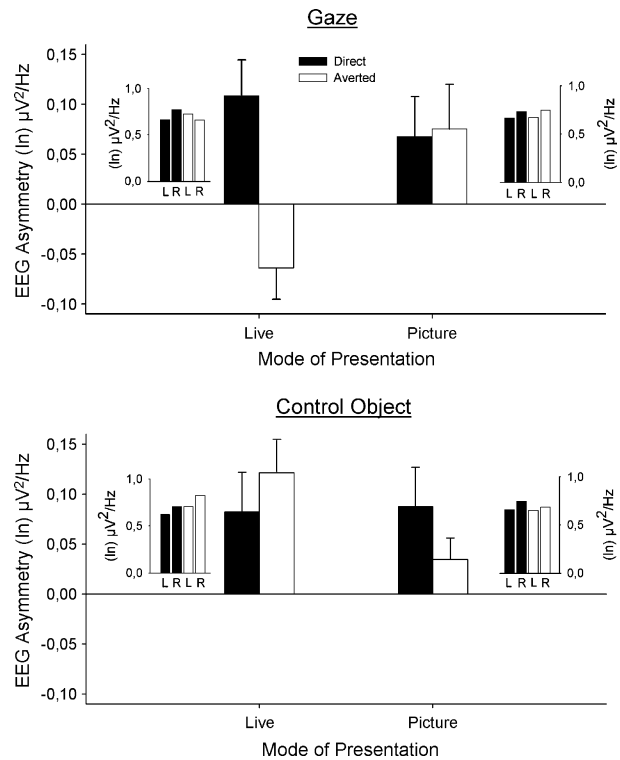

Fig. 2. Mean EEG frontal asymmetry scores for face/gaze and control object as a function of gaze/control object direction and the mode of presentation. The ordinates of the main graphs express the difference in the EEG alpha power between electrodes F4–F3 (in ln-transformed μV^2/Hz). Positive values indicate relative left-sided activation and negative values indicate relative right-sided activation. The small insert graphs show the absolute values of the EEG alpha power recorded from the left hemisphere (L, electrode F3) and right hemisphere (R, electrode F4) electrodes in different stimulus conditions.

Fig. 3 shows the mean SCR (maximum change values) in each condition. A similar ANOVA as above showed that all the main effects (all $ps < .01$) as well as the stimulus × presentation mode, $F(1, 18) = 17.3$, $p < .001$, $\eta_p^2 = .49$, and the stimulus × direction × presentation mode, $F(1, 18) = 4.3$, $p < .05$, $\eta_p^2 = .19$, interactions were significant. Averaged across the stimulus orientation, pairwise comparisons showed that the SCR was significantly larger for the live face than for the other conditions (all $ps < .01$, Bonferroni-corrected). Moreover, in the live face condition, the direct gaze resulted in larger SCR than did the averted gaze, $t(18) = 3.3$, $p < .01$. In the other conditions, the gaze/object direction had no effect on the SCR.

The results from the subjective ratings of arousal and valence are shown in Table 1. For the arousal ratings, the ANOVA indicated that all the main effects and all the two-way interactions were statistically significant (all $ps < .01$). The three-way stimulus × direction × presentation mode interaction was marginally significant, $F(1, 19) = 4.1$, $p < .06$, $\eta_p^2 = .18$. Averaged across the stimulus orientation, pairwise comparisons showed that the arousal ratings were significantly higher for the live face (M live face = 4.4) than for the other conditions (M picture of face = 2.8; M live control object = 2.7; M picture of a control object = 3.0, all $ps < .05$, Bonferroni-corrected). In the live face condition, the direct gaze resulted in higher arousal ratings than did the averted gaze (M direct gaze = 5.4; M averted gaze = 3.3; $t(19) = 5.8$, $p < .001$). The stimulus direction had no effect in the other conditions. For the valence ratings, the ANOVA indicated that the main effect of direc-

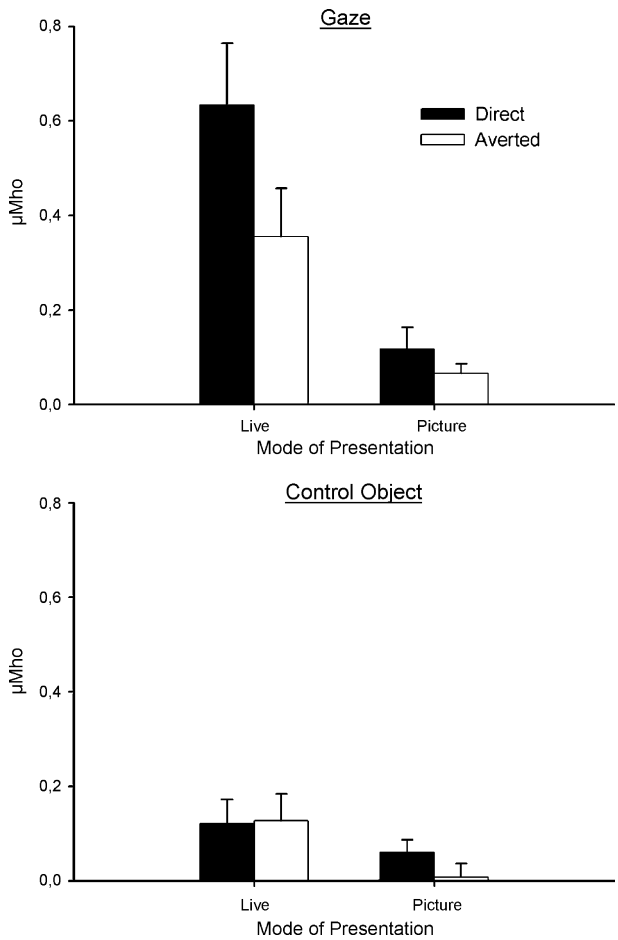

Fig. 3. Mean skin conductance responses (in µMho) for face/gaze and control object as a function of gaze/control object direction and mode of presentation.

tion, $F(1, 19) = 8.4$, $p < .01$, $\eta_p^2 = .31$, and all the interactions were statistically significant (all $ps < .01$). Averaged across the stimulus orientation, pairwise comparisons showed that the valence ratings were significantly more positive for the live control object ($M = 6.6$) than for the picture of a control object ($M = 5.4$), $t(19) = 3.2$, $p < .05$, Bonferroni-corrected. More interestingly, the live faces with an averted gaze direction were rated significantly more pleasant than were the faces with a direct gaze (M averted gaze = 6.8; M direct gaze = 5.4; $t(19) = 4.3$, $p < .001$). In the other conditions, the direction of the gaze/object had no effect on the valence ratings.

3. Discussion

In the present study, we investigated (a) whether seeing another person's direct vs. averted gaze influences the basic motivational–emotional responses of approach and avoidance and (b) whether it would make a difference to these responses if the participants were looking at the face of a real person or a picture. The pictures were presented on a computer monitor and the live stimuli were presented through a liquid crystal shutter. As dependent variables we measured power in the alpha band EEG activity from the left and right frontal channels (F3/F4), skin conductance responses (SCR), and subjective experiences of arousal and emotional valence.

The results provided evidence that seeing a person with a direct vs. averted gaze direction differentially activated the neural approach–avoidance system. However, the mean frontal EEG asymmetry scores differed significantly as a function of gaze direction only when the stimulus was a live face. Pictures of faces with direct and averted gaze or a control object in different orientations (in live or picture conditions) had no effect on the EEG asymmetry. A live direct gaze resulted in relative left-sided frontal activation (indicative of a tendency to approach), whereas a live averted gaze resulted in relative right-sided activation (indicative of avoidance) (cf., Davidson, 1984, 2004; Harmon-Jones, 2003, 2004; Harmon-Jones et al., 2006; Van Honk & Schutter, 2006). Moreover, both of these live gaze direction conditions elicited stronger autonomic responses than did all the other conditions. The measurements of the frontal EEG asymmetry and the SCR demonstrated that another person's gaze direction had an effect on the functioning of the two neural systems postulated to be involved in the regulation of motivational tendencies: one regulating the direction of the responses (i.e., approach or avoidance) and another regulating the intensity of the motivational tendency (Lang et al., 1990). The combination of the frontal EEG asymmetry (approach–avoidance) and the SCR (intensity) values by a multiplicative function, for example, as some theories have suggested (cf., Hull, 1943) would make it very clear that, in the present study, the live direct gaze and averted gaze conditions respectively, triggered stronger motivational tendencies for approach and avoidance than did any of the other conditions.

The subjective evaluations of arousal showed that the live face condition was evaluated as more arousing than were the other conditions and, moreover, that the gaze contact was evaluated as more arousing than the averted gaze. These findings also fit nicely with the present SCR findings. The SCR were also larger to the direct than averted (live) gaze. There is some evidence from earlier SCR studies that the perception of a face elicits a stronger SCR than that elicited by the perception of an object (Hirstein, Iversen, & Ramachandran, 2001) and that eye contact may result in increased SCR as compared with a condition with no eye contact (McBride, King, & James, 1965; Nicholas & Champness, 1971). There are also other earlier studies showing that EEG arousal (decreased alpha activity) and heart rate is higher to eye contact than to averted gaze (Gale, Spratt, Chapman, & Smallbone, 1975; Kleinke & Pohlen, 1971). In the present study, given that the subjective evaluations were performed after the physiological recordings (without the presence of the stimuli), the results suggest that a memory trace was formed of the level of arousal during stimulus presentation, and that it was accessible to the participants when they rated their subjective experiences afterwards. Moreover, the stored informa-

Table 1
Mean ratings for subjective experiences of arousal and valence

Rating	Gaze				Control object			
	Live		Picture		Live		Picture	
	Direct	Averted	Direct	Averted	Direct	Averted	Direct	Averted
Arousal	5.4 (0.34)	3.3 (0.35)	2.9 (0.38)	2.7 (0.36)	2.8 (0.37)	2.7 (0.30)	2.9 (0.34)	3.1 (0.46)
Valence	5.4 (0.42)	6.8 (0.28)	6.2 (0.27)	6.5 (0.26)	6.6 (0.29)	6.6 (0.31)	5.4 (0.41)	5.4 (0.41)

The data are means and standard error of means (in parentheses) for gaze and control object in different conditions of presentation mode and gaze/object direction. For both ratings, the scales range between 1 (calm/unpleasant) and 9 (aroused/pleasant).

tion was accurate enough to allow differentiation in arousal not only between live face and the other stimulus categories, but also between direct and averted gaze of the live face. These findings are interesting when contemplating how people's awareness and memory of their bodily states during social interaction are likely to modulate their future social interactions (cf., Damasio, 1994).

An interesting finding was that the subjective valence ratings indicated that a direct gaze was evaluated as slightly positive, but as less pleasant/positive than was an averted gaze. It is possible that a prolonged exposure (5-s) to another person's direct gaze may have been interpreted as an expression of dominance or aggression (cf., Brooks, Church, & Fraser, 1986) and, therefore, it was evaluated as less positive than a 5-s period of averted gaze. The subjective rating data indicating a less positive response to direct than averted gaze appear to conflict with the EEG asymmetry findings which suggested an approach response to direct gaze. It is important to note, however, that the approach and avoidance responses (as indexed by the EEG asymmetry) are not directly associated with positively and negatively valenced emotional experiences. Motivational approach responses have been demonstrated in the context of both positive and negative emotional experiences (Davidson, 1984; Harmon-Jones, 2003, 2004; Harmon-Jones et al., 2006). The subjective rating data are, therefore, not contradictory to our EEG asymmetry findings, although it is clear that further research is required to examine subjective experiences to direct live gaze in more detail. For example, if it turned out that a prolonged exposure to another's direct gaze elicits negative feelings such as feelings of obtrusiveness or anxiety (associated with stimulus avoidance), the present type of results showing approach-related EEG asymmetry to direct gaze would be difficult to explain. Future research should also determine whether physiological responses and subjective experiences to gaze directions are modulated by other stimulus parameters, e.g., the length of exposure to direct gaze, and factors related to participants' characteristics, e.g., personality traits and cultural conventions regarding gaze behavior.

As already mentioned, the present results showed that the effect of gaze direction was observed only when the participants were facing a real person. When the very same faces were seen as pictures on a computer monitor, there was no difference in the frontal EEG asymmetry between the gaze direction-conditions and the autonomic system was no more strongly activated than in the control conditions. Why would observing a live face vs. a picture of a face elicit differential physiological and behavioral responses? Echoing Argyle (1981) we suggest that the experience of being looked at by a real person initiates "mentalizing" and self-awareness processes differently compared to when facing a picture of another person's face and, furthermore, that these high-level socio-cognitive processes modulate the responses to another person's gaze direction.

To gain support for this suggestion, we measured situational self-awareness from 11 new participants (six females, mean age = 30.6 years, range 24–42 years) while they were facing a real person vs. a picture of a face, both with a direct gaze only. Again, the faces of two females with a neutral expression were used as stimuli and were presented in picture and live conditions using the same apparatus and presentation specifications as in the main experiment. Within each condition (live and picture), the participants saw only three identical 5-s direct gaze trials. While viewing the facial stimuli, the participants were asked to complete a questionnaire containing the Situational Self-Awareness Scale (SSAS, Govern & Marsch, 2001). The SSAS consists of nine items and each item is accompanied by a seven-point scale ranging from 1 (strongly disagree) to 7 (strongly agree). Three of the items measure "public self-awareness" (e.g., Right now, I am concerned about the way I present myself), three items measure "private self-awareness" (e.g., Right now, I am conscious of my inner feelings), and three items measure "awareness of immediate surroundings" (e.g., Right now, I am keenly aware of everything in my environment). It was expected that being a target of another's direct gaze would especially influence the public self-awareness.

Supporting our hypothesis, the results showed that the participants reported significantly elevated levels of public self-awareness in the live than in the picture condition (sum of three ratings measuring public self-awareness, M live = 6.2; M picture = 4.4; $t(10) = 2.3$, $p < .05$). There were no significant differences between the live and picture conditions for the scores on the private self-awareness (M live = 11.5; M picture = 11.0) and awareness of immediate surroundings (M live = 14.2; M picture = 14.4) scales. Thus, the present results suggest that the visual information from perceiving another person with direct and averted gaze may not be enough to activate the approach–avoidance systems. Instead, when the gaze direction is associated with an elevated level of public self-awareness caused, for example, by the awareness of the presence of another person and the feeling of the possibility for being observed, additional socio-cognitive processes are involved and differential patterns of approach–avoidance-related responses to direct and averted gaze are observed.

Indeed, this speculation receives some support from earlier studies measuring SCR to direct and averted gaze. In the early studies reporting greater skin conductance responses to eye contact than to unreciprocated gaze (McBride et al., 1965; Nicholas & Champness, 1971), the stimuli were real persons sitting in front of the participants. However, there are also studies showing no difference in SCR between eye contact and unreciprocated gaze and, in these studies, pictorial stimulus material was used. Leavitt and Donovan (1979) reported that pictures of gazing and non-gazing infants presented on a television monitor did not result in differential skin conductance responses in observing mothers, and Donovan and Leavitt (1980) reported only marginal differences in skin conductance responses between straight gaze and averted head (without eye contact) conditions. We have also recent results from our own laboratory showing no effect of gaze direction on the SCR of typically developing children in a study showing pictorial face stimuli (Kylliäinen & Hietanen, 2006). Thus, the presence of a real person vs. a picture may very well make a difference on the types of responses we were investigating in the present study. Of course, based on the present study, we are not suggesting that pictures of direct and averted gaze *cannot* activate approach–avoidance systems. The present results seem to suggest, however, that potential motivational responses to pictures of faces are not as robust as are responses observed in the presence of a real person.

Earlier studies have shown that direct gaze enhances a variety of social-cognitive functions related to processing of facial information (Macrae et al., 2002; Mason et al., 2004; Vuilleumier et al., 2005). Moreover, direct gaze modulates not only facial perceptions, but it has been shown to increase co-operative and prosocial behavior (Bateson, Nettle, & Roberts, 2006; Haley & Fessler, 2005). It has been suggested that these phenomena may be related to the social and affective significance of direct gaze. These postulations are supported by evidence showing that direct gaze activates the amygdala, a structure having a central role in processing of emotions (George et al., 2001; Kawashima et al., 1999; Wicker et al., 2003). Furthermore, direct gaze has been shown to induce enhanced visual processing of facial information and it is possible that this enhancement reflects the modulation of visual processing exerted by input from the structures (including the amygdala) involved in the processing of emotions (Conty et al., 2007; George et al., 2001; Pelphrey et al., 2004). Interestingly, in all the studies cited above, the reported effects by direct gaze were observed with pictures of faces as stimuli. In the present study, gaze direction in

the pictures of faces had no effect on the measured responses. How can we explain the discrepancy?

An obvious difference is, of course, that in the present study we measured frontal EEG asymmetry and SCRs to direct and averted gaze, whereas, in the studies cited above, the effect of gaze direction was inferred by its influence on other concomitant cognitive processes (behavioral studies) or by its influence on brain responses other than those measured in the present study (imaging and event-related potential studies). However, another possibility relates to the fact that, in the present study, the facial stimuli were presented in the context of passive observation. Instead, in the studies cited above, the effect of direct gaze was demonstrated in the context of a task necessitating discrimination of some sort of facial information (e.g., age, sex, or gaze direction; Conty et al., 2007; George et al., 2001; Macrae et al., 2002; Mason et al., 2004; Kawashima et al., 1999; Pelphrey et al., 2004; Vuilleumier et al., 2005; Wicker et al., 2003) or decision-making, although not related to the faces (Bateson et al., 2006; Haley & Fessler, 2005). It is possible that active engagement by a cognitive task could also provide such a context in which the influence of gaze direction can be observed, even when the face is presented in a pictorial format. We want to emphasize that we are not arguing that increased self-awareness caused by the presence of another person represents the only 'additional socio-cognitive' process which could lead to activation of the approach–avoidance systems in the context of direct and averted gaze. Our results just suggested that when pictures showing an experimenter's face with a neutral expression were presented in a passive observation task, the gaze direction did not activate the approach–avoidance systems.

In the present experiment, two female experimenters posed as stimuli. The inclusion of only two models limits the implications and generalization of the present results. One obvious factor potentially influencing the present type of results relates to the familiarity of the stimulus faces. In this study, the models (experimenters) were not familiar to the participants, but they were not strangers either, as the experimenters and participants had been getting acquainted during the experiment preparation. It is possible that direct gaze by a very familiar vs. strange person would elicit differential responses. Moreover, as the models were the experimenters of the present study, it is possible that they possessed an authoritative status in the eyes of the participants and made the participants to feel being evaluated (cf., Guerin, 1986). This might have contributed, for example, to the present results that the live direct gaze was evaluated as less positive than the averted gaze.

Another potential limitation of the present study relates to the power of the EEG measurements. EEG asymmetry data were obtained from 20 participants by presenting six 5-s trials per stimulus condition. For example, Harmon-Jones et al. (2006) collected data from 55 participants in a paradigm involving sixteen 6-s trials per condition (although data from the first 3 s only were used for the analysis). Now, one could argue, for example, that our data were suffering from poor signal-to-noise ratio and, because of this, we missed an experimental effect in all the other conditions except one (the live face condition). However, we would like to counterargue that the EEG asymmetry effect for gaze direction we found in the live face condition was very clear. Importantly, in the picture condition, there was no indication whatsoever for a gaze direction effect. Moreover, the SCR results were very compatible with the EEG asymmetry results: these effects were also observed in the live face condition only. We think that these features in our results speak against any major problems regarding the signal-to-noise ratio and power of our EEG measurements. Another issue deserving discussion is that our EEG results indicated left-sided asymmetry for all stimulus conditions except one, averted live gaze. Inspection of Fig. 2 reveals that the positive (left-sided) asymmetry score was, in fact, a little higher for 'live averted radio' than it was for 'live direct gaze'. It is a matter of future studies to investigate whether seeing a direct gaze of another person results in more pronounced left-sided frontal EEG asymmetry than seeing of other control stimuli. However, as noted above, when the EEG asymmetry (direction) and SCR (intensity) results are combined, even the present study indicated that seeing a live direct gaze triggered a stronger motivational tendency for approach than did all the other conditions.

Finally, the fact that the present study showed differential physiological responses to live faces vs. pictures of faces is interesting with a view to future work in the area of face perception more generally. Studies with images of faces have provided important results in abundance, and will undoubtedly continue to do so, but the present study shows that there may be instances when seeing a picture of a face and seeing a real face result in differing findings. In fact, a recent study from our laboratory using the same methodology as in the present study showed differences in the event-related potential (ERP) responses to a neutral face of a real person and a realistic dummy (both with direct gaze) when these stimuli were presented live but not when presented as pictures (Pönkänen et al., 2008). Thus, it is possible that some effects on physiological responses may not be observed at all when face processing is investigated using pictures of faces, or that the effects are diminished as compared with effects elicited by real faces. Future studies will hopefully shed light upon these issues.

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